23MIC0055 KISHORI KAMALI M

In [1]:
```
import pandas as pd
```

a. Create a simple series of Student_Names and Programmes they belong with

In [2]:
```
ss=pd.Series(["kishori","rajavel","nideesh"],index=["MIC","MID","MIC"])
print(ss)
```

```
MIC    kishori
MID    rajavel
MIC    nideesh
dtype: object
```

b. Create a series from a list, tuple, dictionary and a custom index

In [5]:
```
lst=[90,98,100]
print(f"the type of data is {type(lst)}")
lss=pd.Series(lst)
print(lss)
```

```
the type of data is <class 'list'>
0     90
1     98
2    100
dtype: int64
```

In [6]:
```
tup=(20,30,40)
print(f"the type of data is {type(tup)}")
tss=pd.Series(tup)
print(tss)
```

```
the type of data is <class 'tuple'>
0    20
1    30
2    40
dtype: int64
```

In [8]:
```
dic={"kishori":99,"rajavel":100,"nideesh":98}
print(f"the type of data is {type(dic)}")
dss=pd.Series(dic)
print(dss)
```

```
the type of data is <class 'dict'>
kishori     99
rajavel    100
nideesh     98
dtype: int64
```

In [9]:
```
cust=pd.Series(["best","better","good"],index=["kishori","nideesh","rajavel"])
print(cust)
```

```
kishori      best
nideesh    better
rajavel      good
dtype: object
```

c. Create two series with CAT I and CAT II marks of students and explore element-wise operations

In [28]:
```
s1=pd.Series([90,98,99],index=["python","textmining","nlp"])
s2=pd.Series([100,98,99],index=["python","textmining","nlp"])
print(s1+s2)
print("\n")
print(s1["python"]*s2["textmining"])
print("\n")
print(s1*s2)
print("\n")
print((s1+s2).mean())
```

```
python        190
textmining    196
nlp           198
dtype: int64


8820


python        9000
textmining    9604
nlp           9801
dtype: int64


194.66666666666666
```

d. Write conditions to check if data in a series are integers/chars/Not a Number

In [42]:
```
ss=pd.Series([10,20,30],index=["m1","m2","m3"])
print(ss.isna())
print(ss.isnull())
```

```
m1    False
m2    False
m3    False
dtype: bool
m1    False
m2    False
m3    False
dtype: bool
```

In [43]:
```
sem=[["cat1","cat1","cat1","cat2","cat2","cat2"],["python","textmining","nlp","python","textmining","nlp"]]
index=pd.MultiIndex.from_arrays(sem,names=["exam","subject"])
student=pd.Series([90,98,99,100,98,99],index=index)
print(student)
print("sucessfully created")
```

```
exam  subject
cat1  python         90
      textmining     98
      nlp            99
cat2  python        100
      textmining     98
      nlp            99
dtype: int64
sucessfully created
```

e. Create a series for Product Names, Prices, Quantities in a retail store

In [55]:
```
ss=pd.Series(["waterbottle","milton",120,60],index=["product_name","brand_name","price","quantity"])
print(ss)
print(type(ss["product_name"]))
print(type(ss["price"]))
```

```
product_name    waterbottle
brand_name           milton
price                   120
quantity                 60
dtype: object
<class 'str'>
<class 'int'>
```

In [57]:
```
p=[["waterbottle","clutch","fakenails"],[100,50,200]]
index=pd.MultiIndex.from_arrays(p,names=["product_names","prices"])
ss=pd.Series([1,5,10],index=index)
print(ss)
```

```
product_names  prices
waterbottle    100        1
clutch         50         5
fakenails      200       10
dtype: int64
```

f. Create a Multilevel series with arrays, tuple, product and dataframe

created fro tuples

In [10]:
```
import pandas as pd
tup=(("kishori",99),("rajavel",98))
index=pd.MultiIndex.from_tuples(tup,names=["student_name","mark"])
n=pd.Series(["good","better"],index=index)
print("created from tuple\n")
print(n)
```

```
created from tuple

student_name  mark
kishori       99        good
rajavel       98      better
dtype: object
```

created from data frame

In [30]:
```
import pandas as pd
data={"student_name":["kishori","rajavel"],
      "marks":[100,100]}
df=pd.DataFrame(data)
index=pd.MultiIndex.from_frame(df)
n=pd.Series(["good","better"],index=index)
print("created from data frame\n")
print(n)
```

```
created from data frame

student_name  marks
kishori       100        good
rajavel       100      better
dtype: object
```

created from arrays

In [16]:
```
p=[["waterbottle","clutch","fakenails"],[100,50,200]]
index=pd.MultiIndex.from_arrays(p,names=["product_names","prices"])
ss=pd.Series([1,5,10],index=index)
print(ss)
```

```
product_names  prices
waterbottle    100        1
clutch         50         5
fakenails      200       10
dtype: int64
```

created from products

In [29]:
```
d1=[[1,2,3],["bottle","clucth","noodles"]]
index=pd.MultiIndex.from_product(d1)
ss=pd.Series([100,200,300,400,500,600,700,800,900],index=index)
print(ss)
```

```
1  bottle     100
   clucth     200
   noodles    300
2  bottle     400
   clucth     500
   noodles    600
3  bottle     700
   clucth     800
   noodles    900
dtype: int64
```

g. Access the data elements using First Level Index and Second Level Index.

In [19]:
```
sem=[["cat1","cat1","cat1","cat2","cat2","cat2"],["python","textmining","nlp","python","textmining","nlp"]]
index=pd.MultiIndex.from_arrays(sem,names=["exam","subject"])
student=pd.Series([90,98,99,100,98,99],index=index)
print(student)
print("sucessfully created")
```

```
exam  subject
cat1  python         90
      textmining     98
      nlp            99
cat2  python        100
      textmining     98
      nlp            99
dtype: int64
sucessfully created
```

In [20]:
```
print(student["cat1"])
```

```
subject
python        90
textmining    98
nlp           99
dtype: int64
```

In [21]:
```
print(student["cat1","python"])
```

```
90
```

h. Access a specific data item using .loc

first level retrieval using loc

In [22]:
```
print(student.loc["cat2"])
```

```
subject
python        100
textmining     98
nlp            99
dtype: int64
```

slicing using loc

In [23]:
```
print(student.loc["cat1":"cat2"])
```

```
exam  subject
cat1  python         90
      textmining     98
      nlp            99
cat2  python        100
      textmining     98
      nlp            99
dtype: int64
```

specific data using loc/second level retrieval using loc

In [26]:
```
print(student.loc["cat1","textmining"])
```

```
98
```

In [ ]:

In [ ]: